CSE 464: Software QA and Testing

Project Part #2 — README

Amir Boshra

Arizona State University

Spring 2026

GitHub Repository: https://github.com/aboshra11/CSE-464-2026-aboshra

1. How to Compile and Run

Prerequisites: Java JDK 21, Maven, Graphviz must be installed and on PATH.

Compile & Run All Tests

```
mvn package
```

This compiles the project and runs all JUnit tests automatically. You should see BUILD SUCCESS with all 13 tests passing.

Run the Main Class

```
java -cp target/CSE464-2026-aboshra-1.0-SNAPSHOT.jar edu.asu.cse464.Main
```

2. Feature 1 — Remove Node and Edge APIs

Three new APIs were added to Graph.java to support removing nodes and edges from the graph.

2.1 removeNode(String label)

Removes a single node by label. Also removes all edges connected to that node. Throws IllegalArgumentException if the node does not exist.

Example:

```
graph.addEdge("a", "b");
graph.addEdge("b", "c");
graph.removeNode("b"); // removes node b and edges a->b and b->c
```

2.2 removeNodes(String[] labels)

Removes multiple nodes at once by iterating and calling removeNode() for each. Throws IllegalArgumentException if any node does not exist.

Example:

```
graph.removeNodes(new String[]{"a", "b"});
```

2.3 removeEdge(String srcLabel, String dstLabel)

Removes a directed edge between two nodes. Nodes remain in the graph. Throws IllegalArgumentException if either node or the edge does not exist.

Example:

```
graph.addEdge("a", "b");
graph.removeEdge("a", "b"); // removes edge, nodes a and b remain
```

2.4 Test Cases

Tests are in GraphTest.java and cover the 3 required scenarios:

- Scenario 1: Nodes and edges are correctly removed (testRemoveNode, testRemoveNodes, testRemoveEdge)
- Scenario 2: Removing a node that does not exist throws IllegalArgumentException (testRemoveNodeNotFound, testRemoveNodesNotFound)
- Scenario 3: Removing an edge that does not exist throws IllegalArgumentException (testRemoveEdgeNotFound, testRemoveEdgeNodeNotFound)

```
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< edu.asu:CSE464-2026-aboshra >---------------------
[INFO] Building CSE464-2026-aboshra 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ CSE464-2026-aboshra ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ CSE464-2026-aboshra ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ CSE464-2026-aboshra ---
[INFO] skip non existing resourceDirectory C:\Users\Dell\IdeaProjects\CSE464-GraphParser\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ CSE464-2026-aboshra ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ CSE464-2026-aboshra ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running edu.asu.cse464.GraphTest
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.078 s -- in edu.asu.cse464.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ CSE464-2026-aboshra ---
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.196 s
[INFO] Finished at: 2026-04-03T17:27:11-07:00
```

2.5 Commit Link

- **Remove APIs commit:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/0bc9e71

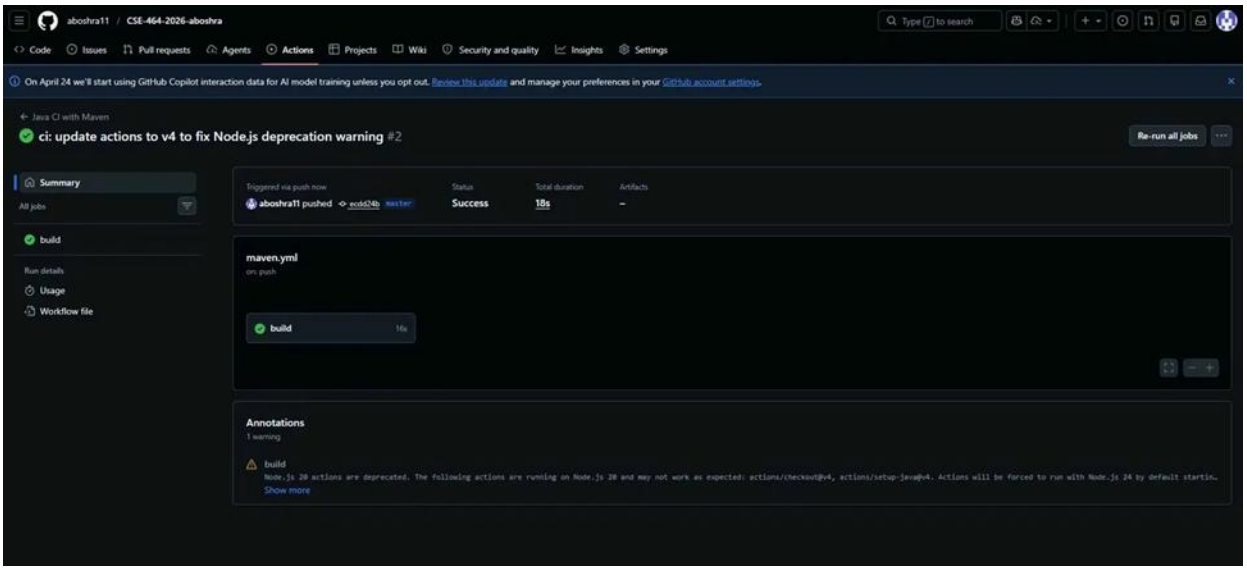
3. Continuous Integration (GitHub Actions)

A GitHub Actions workflow was added at .github/workflows/maven.yml. Every push to master automatically triggers a build and runs all tests using mvn package.

Workflow File

```
name: Java CI with Maven
on:
  push:
    branches: [ "master" ]
  pull_request:
```

```
    branches: [ "master" ]
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-java@v4
        with:
          java-version: '21'
          distribution: 'temurin'
      - run: mvn package
```



Commit Links

- **CI workflow added:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/cbee066
- **CI updated to v4 actions:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/ecdd24b

4. BFS Branch — GraphSearch with BFS

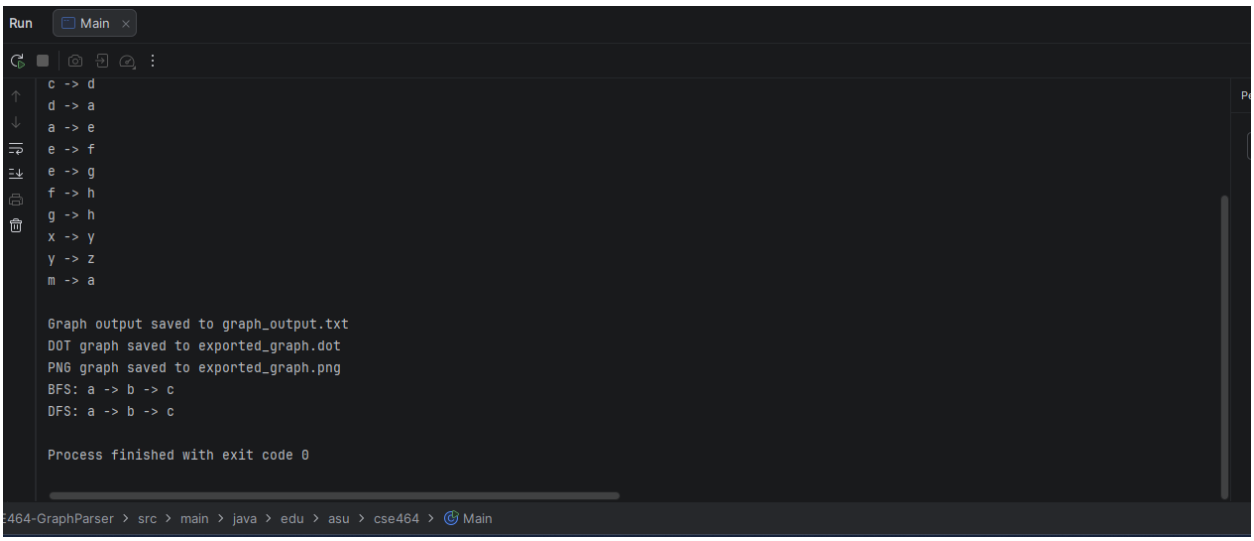
A new branch called bfs was created. The GraphSearch(Node src, Node dst) API was implemented using Breadth-First Search (BFS).

New Classes

- Path.java — represents a path as a list of nodes, printed as: n1 -> n2 -> n3
- GraphSearch(Node src, Node dst) added to Graph.java — returns a Path if one exists, null otherwise

Example Usage

```
graph.addEdge("a", "b");
graph.addEdge("b", "c");
Node src = graph.getNodes().get("a");
Node dst = graph.getNodes().get("c");
Path path = graph.GraphSearch(src, dst); // BFS
System.out.println(path); // a -> b -> c
```



Branch & Commit Links

- **bfs branch:** https://github.com/aboshra11/CSE-464-2026-aboshra/tree/bfs
- **BFS commit:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/df9e254

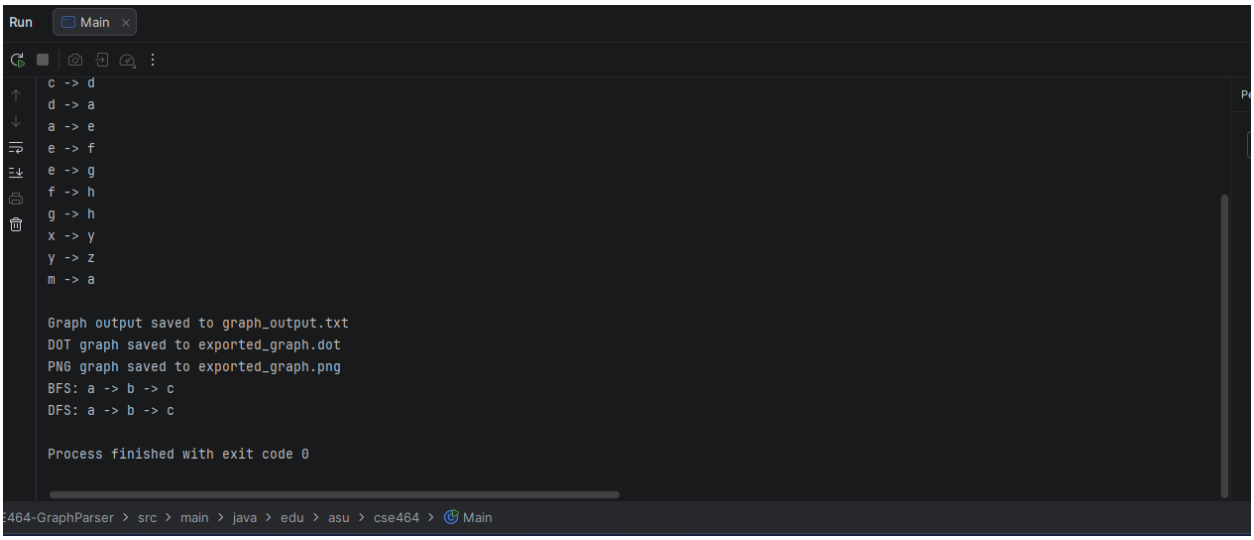
5. DFS Branch — GraphSearch with DFS

A new branch called dfs was created. The same GraphSearch(Node src, Node dst) API was implemented using Depth-First Search (DFS).

Example Usage

```
graph.addEdge("a", "b");
graph.addEdge("b", "c");
Node src = graph.getNodes().get("a");
Node dst = graph.getNodes().get("c");
```

```
Path path = graph.GraphSearch(src, dst); // DFS
System.out.println(path); // a -> b -> c
```



Branch & Commit Links

- **dfs branch:** https://github.com/aboshra11/CSE-464-2026-aboshra/tree/dfs
- **DFS commit:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/d17b2c3

6. Merge Conflict Resolution — Algorithm Enum

The bfs branch was merged into master first. Then the dfs branch was merged, causing a conflict in Graph.java since both branches defined GraphSearch(Node src, Node dst). The conflict was resolved by combining both algorithms into a single method with an Algorithm enum parameter.

Final API

```
public Path GraphSearch(Node src, Node dst, Algorithm algo)
```

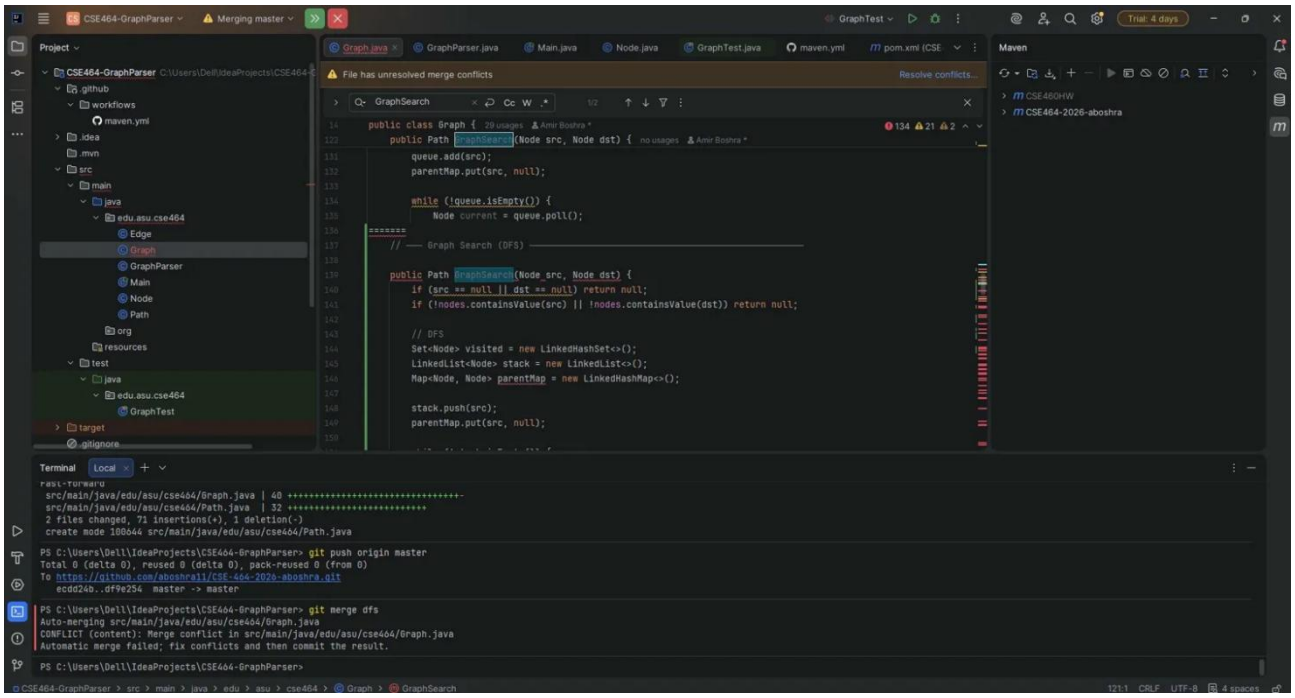
Algorithm Enum

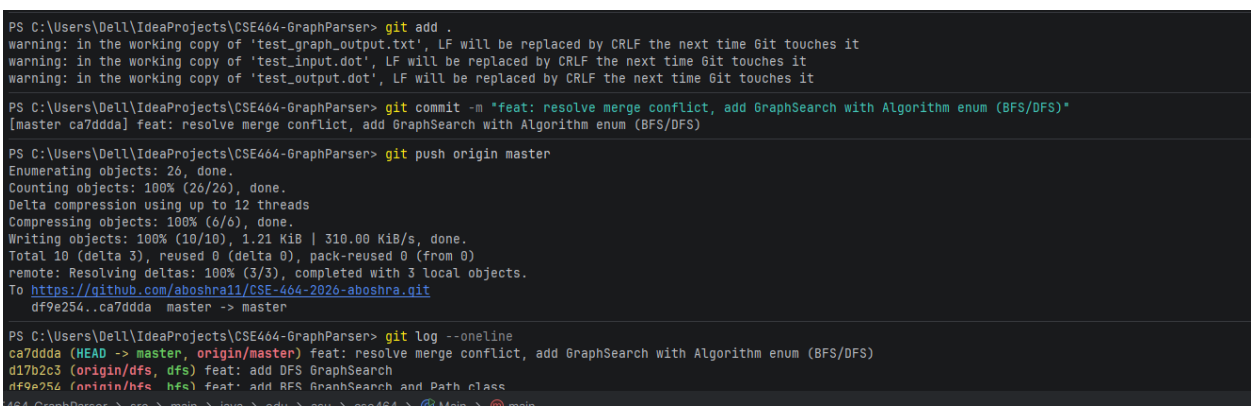
```
public enum Algorithm {
    BFS,
    DFS
}
```

Example Usage

```
Node src = graph.getNodes().get("a");
```

```
Node dst = graph.getNodes().get("c");
Path bfsPath = graph.GraphSearch(src, dst, Algorithm.BFS);
Path dfsPath = graph.GraphSearch(src, dst, Algorithm.DFS);
System.out.println(bfsPath); // a -> b -> c
System.out.println(dfsPath); // a -> b -> c
```





Commit Link

- **Merge conflict resolution commit:** https://github.com/aboshra11/CSE-464-2026-aboshra/commit/ca7ddda

7. Summary of All Commits and Branches

Feature	Commit / Branch	Link
Remove APIs	`0bc9e71`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/0bc9e71
CI Workflow	`cbee066`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/cbee066
CI v4 Fix	`ecdd24b`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/ecdd24b
BFS branch	`bfs`	https://github.com/aboshra11/CSE-464-2026-aboshra/tree/bfs
BFS commit	`df9e254`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/df9e254
DFS branch	`dfs`	https://github.com/aboshra11/CSE-464-2026-aboshra/tree/dfs
DFS commit	`d17b2c3`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/d17b2c3
Merge + Enum	`ca7ddda`	https://github.com/aboshra11/CSE-464-2026-aboshra/commit/ca7ddda